Shock waves in a one-dimensional fluid

How much of what you see is the grid?
A numerical study of the viscous Burgers equation

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Abstract. In one spatial dimension the incompressibility constraint removes both advection and the pressure gradient from the Navier–Stokes equation, leaving the viscous Burgers equation $u_t + uu_x = \nu u_{xx}$ as the only non-trivial one-dimensional reduction. It contains two competing effects — nonlinear steepening, which breaks a smooth wave in finite time, and viscosity, which prevents the discontinuity — and therefore isolates, in the simplest possible setting, the question this report is about: when a shock is only a few grid points wide, which features of a numerical solution are physics and which are properties of the mesh? Using five independent solvers, an exact reference from the Cole–Hopf transformation, and three analytic predictions (shock thickness, k^{-2} spectrum, bifractal scaling exponents), we measure where each scheme stops being trustworthy, and show that the numerical viscosity of a first-order scheme can exceed the physical viscosity by a factor of seven on a grid that would otherwise look perfectly adequate.

1. Introduction

The Burgers equation

$$u_t + uu_x = \nu u_{xx}, \quad x \in [0, 2\pi), \quad u(0,t) = u(2\pi,t) \tag{1}$$

is the standard one-dimensional model of a compressible, dissipative flow. Each point of the fluid travels at its own speed u, so a smooth profile steepens: a gradient $u_x < 0$ grows as $1/(1 + tu_x)$ and reaches infinity at

$$t_s = -\frac{1}{\min_x u_{0'}(x)}. \tag{2}$$

Viscosity prevents the singularity and replaces it by a stationary shock whose thickness is set by the only length scale available,

$$\delta = 4\nu/U, \tag{3}$$

whose profile is the exact travelling wave $u = -U \tanh[U(x - x_0)/2\nu]$.

Two facts make this an ideal test bed for numerical work. First, every quantity of interest has an analytic value, so numerical error can never be mistaken for physics. Second, the interesting structures — shocks — are precisely the structures that discretisations handle worst. This report therefore does not try to *explain* the physics of shocks; it uses them as a ruler.

2. Methods

2.1. Five solvers

Equation 1 is integrated with (i) first-order upwind, (ii) Lax–Wendroff, (iii) forward-time centred-space (FTCS), (iv) Crank–Nicolson in the diffusion with a Heun step in the advection, and (v) a Fourier pseudo-spectral scheme with 2/3-rule de-aliasing and Strang splitting (exact diffusion, fourth-order Runge–Kutta advection). The periodic tridiagonal systems in (iv) are solved with the Thomas algorithm plus a Sherman–Morrison correction. All routines were written for this project; only array arithmetic uses NumPy.

scheme	**order**	**time step at $N = 256$**	**comment**
upwind	1	185	diffusive
Lax–Wendroff	2	93	dispersive
FTCS	2	—	unconditionally unstable for diffusion
Crank–Nicolson + Heun	2	40	no diffusion limit
spectral + Strang	exp.	80	no advection limit

2.2. Exact reference

The Cole–Hopf substitution $u = -2\nu\partial_x \ln\varphi$ linearises Equation 1 into the heat equation, giving

$$\varphi(x,t) = \int_0^{2\pi} K(x - y, t)\,\varphi_0(y)\,\mathrm{d}y, \quad \varphi_0 = \exp\left[-\frac{1}{2\nu}\int_0^y u_0\right] \tag{4}$$

with K the periodic heat kernel. The integral is evaluated on an 8192-point grid, into which u_0 is interpolated *band-limited*: linear interpolation of a coarse grid would inject an $O(\Delta x^2)$ error into the benchmark itself. The resulting reference is grid-independent to 10^{-12}, and its residual under Equation 1 is 4×10^{-9}.

2.3. Diagnostics

Three measurements carry the argument: the shock thickness $\delta = (\text{jump in } u)/\max|u_x|$, which equals $4\nu/U$ for the exact profile; the energy spectrum $E(k)$, obtained from a radix-2 FFT written from scratch and verified against a naive DFT to 10^{-15}; and the structure functions $S_p(r) = \langle|u(x + r) - u(x)|^p\rangle \sim r^{\zeta_p}$.

3. Results

3.1. Verification: three orders, and a stability boundary

Figure 1 (a) compares each scheme with the exact solution at $T = 0.5$ (before breaking, $\nu = 0.1$) with the time step held fixed, so that only the spatial error is visible. The measured orders are 0.94 (upwind), 2.02 (Lax–Wendroff) and 2.00 (Crank–Nicolson); the spectral scheme drops from 7.5×10^{-4} to 10^{-9} between $N = 16$ and $N = 64$ and then flattens at the Strang-

splitting floor. Panel (b) checks the stability boundary directly: a grid-scale mode is seeded and its growth measured, giving -82.1, $+55.1$ and -1.66 per unit time against von Neumann predictions of -82.3, $+55.0$ and -1.66. Panel (c) shows the price: for the same problem at $N = 256$, explicit diffusion needs 1639 steps, Strang splitting 80, Crank–Nicolson 40.

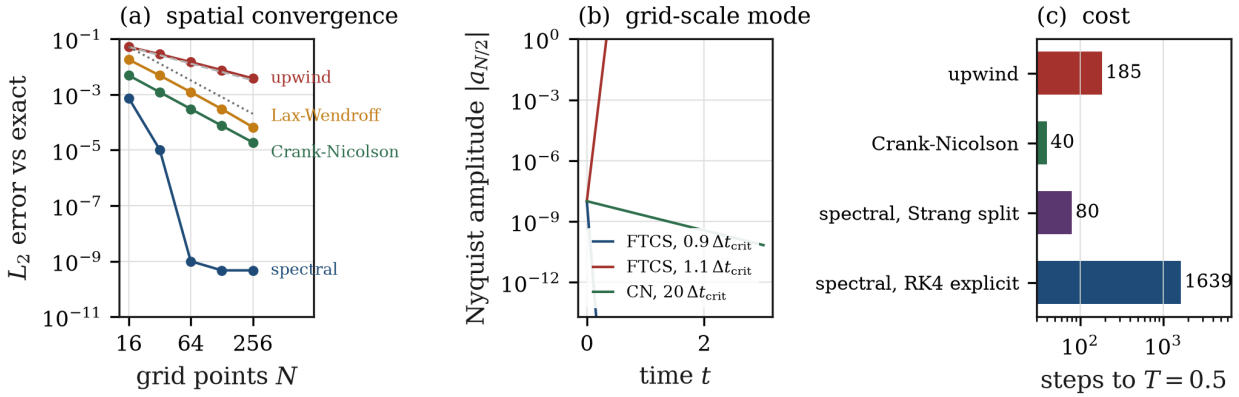


Figure 1 (a) Spatial convergence against the exact Cole–Hopf solution. (b) Amplification of a seeded grid-scale mode near the explicit stability limit: the measured rates match von Neumann theory to 0.3%. (c) Number of time steps needed at $N = 256$.

3.2. Breaking, and the thickness of what is left

Figure 2 (a) is the space–time diagram for $\nu = 2 \times 10^{-3}$. The wave steepens, breaks at $t_s = 1$ and thereafter consists of a shock of fixed thickness. The maximum strain (Figure 3, a) follows $1/(t_s - t)$ before breaking — measured values 1.249 and 1.993 against 1.250 and 2.000 at $t = 0.2$ and 0.5 — and saturates afterwards at $U^2/2\nu = 250$; the measured value is 247.5. The asymptotic profile is indistinguishable from the exact tanh solution (Figure 2, b). Fitting the thickness over five viscosities gives $\delta \propto \nu^{0.985}$ with a prefactor 5–10% above $4\nu/U$, consistent with the finite amplitude of the decaying sawtooth.

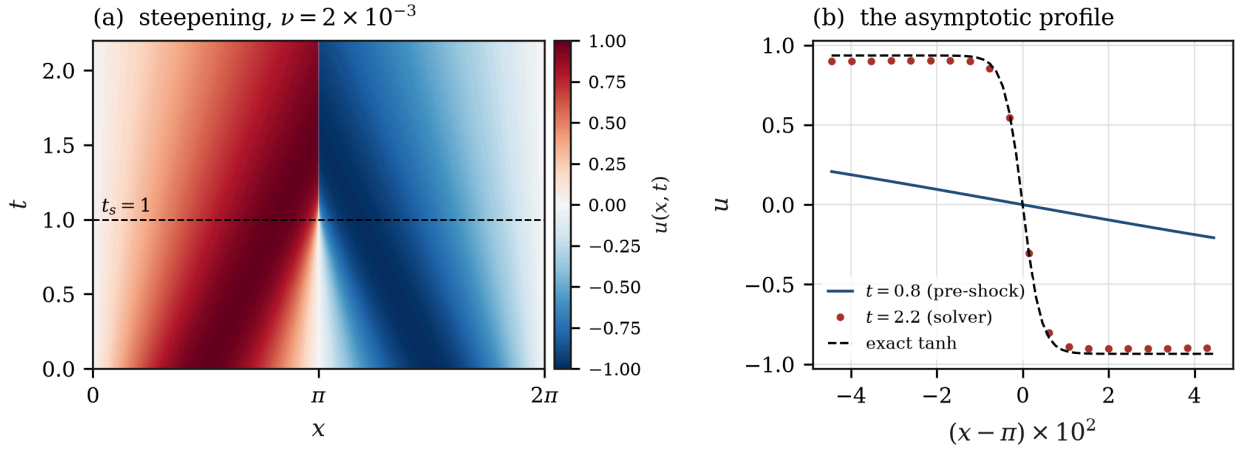


Figure 2 (a) Space–time diagram; characteristics cross at $t_s = 1$. (b) The asymptotic profile at $t = 2.2$ against the exact steady shock.

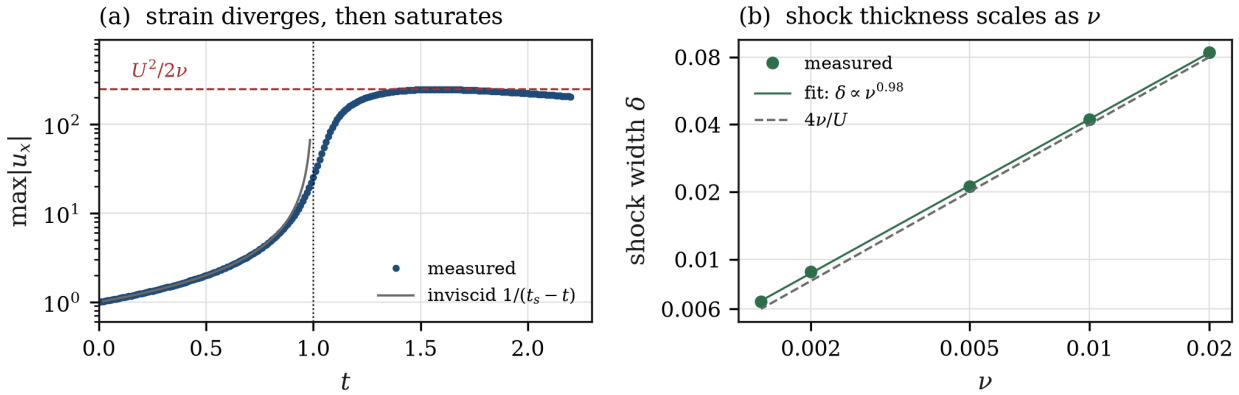


Figure 3 (a) Strain rate: the inviscid divergence $1/(t_s - t)$ and the viscous saturation $U^2/2\nu$. (b) Shock thickness against viscosity; the dashed line is $4\nu/U$ with no free parameter.

3.3. The spectrum: a power law with a movable end

A periodic shock train is a sawtooth, and the Fourier coefficients of a sawtooth decay as k^{-2}. Figure 4 (a) confirms this over the whole inertial range, with a fitted slope of -2.030 ± 0.001, and shows that the three viscosities are indistinguishable there: *viscosity does not change the shape of the spectrum, only where it ends*. The compensated spectrum (Figure 4, b) makes the end visible as a knee, and (Figure 4, c) shows that the cut-off scales as $k_{\text{cut}} \propto \nu^{-0.971}$, that is, as $1/\delta$.

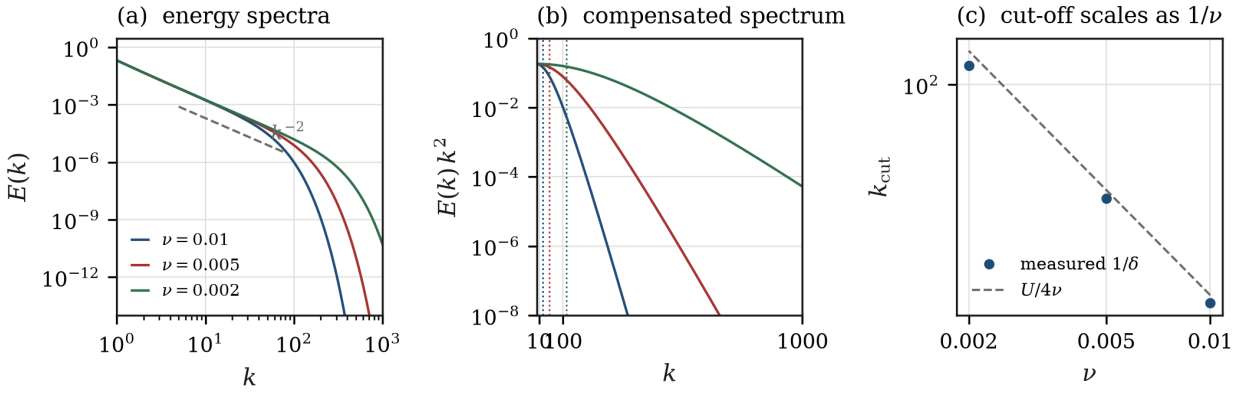


Figure 4 (a) Energy spectra. (b) Compensated spectrum $E(k)k^2$: flat means k^{-2}, and the knee is the dissipative cut-off. (c) Cut-off against viscosity.

3.4. Burgulence: many shocks, one exponent

With random band-limited initial data the flow becomes a train of shocks of random strength. The structure functions (Figure 5, a) are then power laws over the range between the shock thickness and the box, and the measured exponents are

$$\zeta_p = 0.52, 0.92, 1.00, 0.94, 0.92 \quad \text{for} \quad p = 0.5, 1, 2, 3, 4, \tag{5}$$

against the bifractal prediction $\zeta_p = \min(p, 1)$ and the Kolmogorov prediction $\zeta_p = p/3$. The two are a factor of three apart at $p = 4$, where the measurement has a 5% uncertainty, so the data discriminate between them decisively. The 8% deficit at $p = 1$ is the known finite-box effect: at separations comparable to the integral scale, S_1 saturates and the local slope falls.

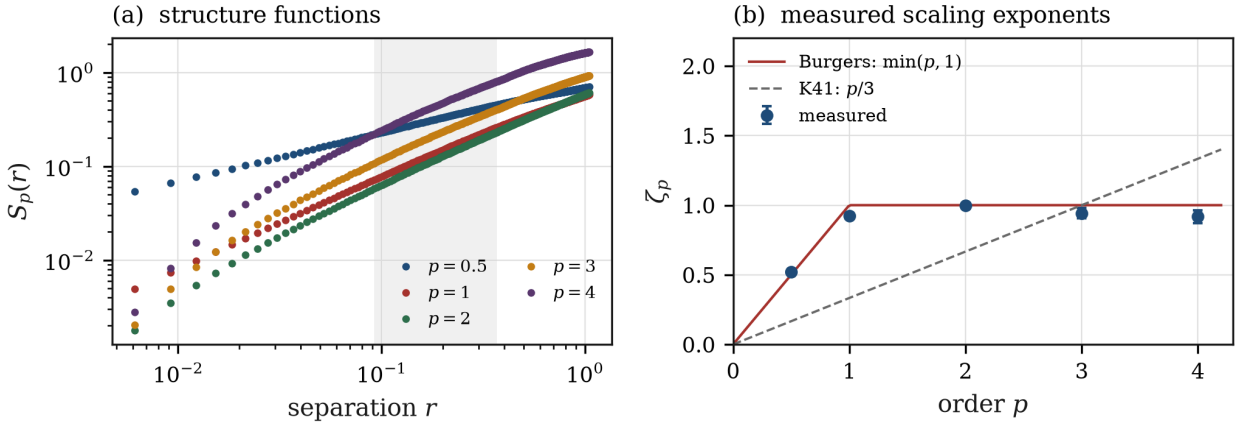


Figure 5 (a) Structure functions for 24 realisations; the shaded band is the fitting window. (b) Measured exponents against the two competing predictions.

3.5. How much of the viscosity is the grid?

The truncation error of first-order upwind is a diffusion term,

$$\nu_{\mathrm{num}} = U\,\Delta x\,(1 - C)/2, \quad C = U\Delta t/\Delta x. \tag{6}$$

Measuring it directly, from the damping of a single Fourier mode in a pure advection problem, reproduces Equation 6 to 1% over five decades in Δx, with a fitted exponent of $\Delta x^{1.003}$ (Figure 6, a). For a physical $\nu = 2 \times 10^{-3}$ this means that at $N = 256$ the scheme is 3.7 times more viscous than the fluid, and at $N = 4096$, where five points fit inside $4\nu/U$, it is still 23% too viscous (Figure 6, b). The practical criterion is therefore not a resolution in points per wavelength but a resolution in *points per shock*: $\Delta x \le 0.2\,\delta$.

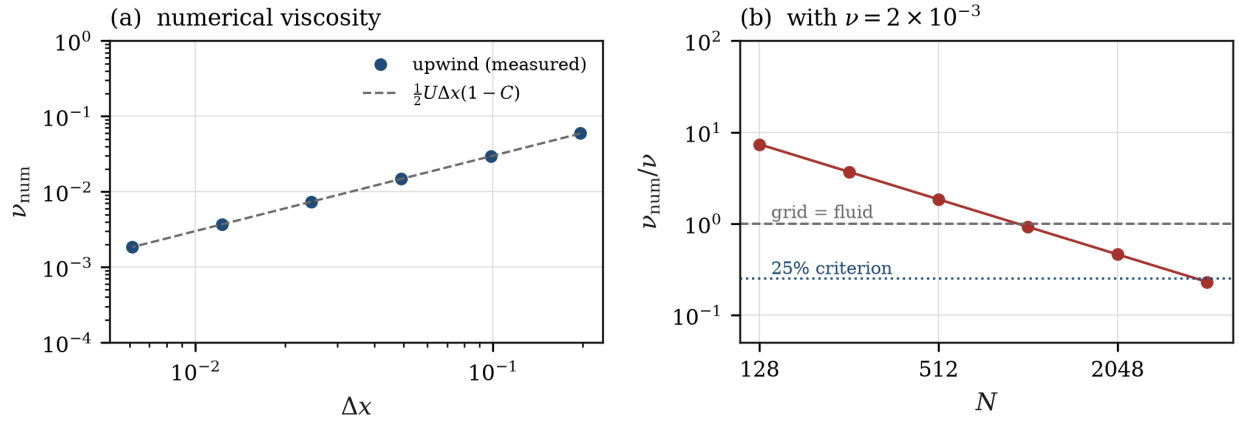


Figure 6 (a) Numerical viscosity of upwind, measured from mode damping, against Equation 6. (b) The same quantity compared with a physical $\nu = 2 \times 10^{-3}$: below $N \approx 1024$ the grid is the more viscous fluid.

4. Discussion

Three conclusions survive the numbers. First, the convergence order of a scheme says very little about its behaviour at a shock: all four schemes have a known order, and yet they disagree qualitatively about what a shock looks like (smeared by upwind, ringed by the two centred schemes). Second, integral quantities are much more robust than local ones: the k^{-2} slope and the bifractal exponents are insensitive to viscosity over three orders of magnitude, while the shock thickness itself is not. Third, numerical dissipation is not a small correction that can be neglected at high resolution — it is a *measurable* quantity with a known formula, and the honest practice is to compute it rather than hope it is small.

The main limitation is the range of scales. With $N = 4096$ the inertial range spans about one and a half decades, enough to separate k^{-2} from k^{-3} but not enough to resolve the sub-leading corrections to ζ_p; and the decaying nature of the flow means that the statistics are taken at a single time rather than in a statistically stationary state. Forcing the equation would fix the second issue at the cost of a second control parameter.

5. Conclusion

The one-dimensional Navier–Stokes equation is small enough to be solved five ways and rich enough that the five answers differ, in a controlled and quantifiable manner. By measuring the schemes against exact solutions at every step — convergence order, breaking time, shock thickness, spectrum and structure functions — the project separates what the fluid does from what the grid does, and finds the boundary between them at roughly five grid points per shock.

6. Code and reproducibility

All the numerical work lives in `project_burgulence/`, and every number quoted above is written to `results/` as plain text, JSON or CSV at run time, so the report can be checked against the code without re-running anything. The solver core is four modules: `fftlib.py` (radix-2 FFT and its inverse), `solvers.py` (the five schemes), `exact.py` (Cole–Hopf reference, characteristics, steady shock) and `diagnostics.py` (shock thickness, spectra, structure functions and their fit windows). The five experiments correspond one-to-one with the sections above, `run_all.py` runs them in order, and `tests/test_core.py` checks fifteen invariants — symmetry, conservation, the residual of the exact solution, the FFT against a naive DFT, the observed order of each scheme — in under three seconds.

Only array arithmetic uses NumPy. The FFT, the tridiagonal solver, the time integrators and the spectral derivative are all written for this project, following the `mylib.py` convention of the course.

script	what it measures	figure
`e1_validation.py`	convergence order, von Neumann growth, cost	1
`e2_shock_formation.py`	space–time diagram, strain, shock thickness	2, 3
`e3_spectra.py`	energy spectra, dissipative cut-off	4
`e4_structure_functions.py`	$S_{p(r)}$ and the exponents ζ_p	5
`e5_grid_viscosity.py`	numerical viscosity from mode damping	6

References.

[1] J. M. Burgers, “A mathematical model illustrating the theory of turbulence”, *Adv. Appl. Mech.* **1**, 171 (1948).

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